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PRIVACY, SECURITY, AND DATA GOVERNANCE

A. Overview

In various embodiments, the disclosed system is configured to support privacy-preserving measurement, processing, and presentation of intimate physiology. The system may be configured to reduce identifiability, restrict sharing of raw signals, and provide user-controlled permissions for any cross-device and cross-user correlation features.

B. Data Minimization and Abstraction

In certain embodiments, the system minimizes storage and sharing of raw signals by transforming raw sensor outputs into derived features and privacy-preserving metrics. In some embodiments, raw signals are stored only temporarily, stored locally on a node or mobile device, or excluded entirely from cross-user workflows. In various embodiments, the system outputs abstract indices, ranges, zones, curves, and confidence indicators rather than explicit anatomical imagery or explicit anatomical labels.

C. Access Controls and Consent

In various embodiments, access to user data and correlation features is governed by user authorization. Authorization may include one or more of explicit consent prompts, revocable permissions, time-limited sharing, one-way or two-way sharing selection, pairing invitations, and consent scopes limiting which data types may be shared. In certain embodiments, cross-user correlation is disabled by default and requires affirmative consent by each participating user.

D. Encryption and Secure Transport

In various embodiments, data is encrypted in transit and at rest. Data transport between nodes and a computing system may use authenticated communication and encryption. Storage on a mobile device and/or remote server may include encryption at rest, key management procedures, and access controls. In certain embodiments, the system supports secure pairing of nodes and prevents unauthorized devices from accessing node data.

E. Anonymization and Privacy-Preserving Transformations

In certain embodiments, the system applies transformations to reduce identifiability while preserving correlation utility. Such transformations may include normalization, quantization, hashing, compression, dimensionality reduction, learned embeddings, or other representations. In some embodiments, the system uses privacy-preserving exchange of derived metrics such that raw signals and/or sensitive raw features are not shared between users.

F. Auditability, Revocation, and Expiration

In various embodiments, the system supports revocation of permissions and expiration of sharing sessions. In certain embodiments, users may view and modify sharing settings, revoke access, and delete shared tokens. In some embodiments, the system maintains audit logs indicating when correlation was performed and which data categories were used, without exposing underlying raw data.

G. Processing Locations and Modes

In certain embodiments, processing is performed locally on-node and/or on a user device. In other embodiments, processing is performed in a cloud environment with access controls. The system may support hybrid modes in which sensitive transformations are performed locally while less sensitive computations occur remotely. In some embodiments, users may select privacy modes that determine whether processing occurs locally, remotely, or in a mixed configuration.

H. Variations

Any privacy and security approach described herein may be combined with other approaches described herein. The system may implement additional safeguards including rate limiting, anomaly detection, device attestation, and spoof-resistance checks. The described privacy and security features may be applied across all nodes, all workflows, and all pairing contexts.